Name: Priyanshi Singh
Class: T22
Roll No: 2401115

ASD&D EXPERIMENT 6

Aim: To study and Implement Agile Software Project lifecycle using Jira with DevOps Pipeline Integration._

LO mapped: LO1

LO: Apply Agile methodologies and use Jira for Sprint Planning, task tracking and integrating with DevOps workflow.

Theory:

What is Jira?

Jira is an Agile project management and issue tracking tool developed by Atlassian. It helps software engineering teams plan sprints, track tasks, manage bugs, and monitor releases across customizable workflows.

Core Concepts & Hierarchy

- Project: A shared workspace representing a product, application, or team initiative.
- Issue Hierarchy:
  - Epic: A large objective or high-level feature broken down into smaller tasks (e.g., *Core Infrastructure*).
  - Story: An end-user feature written from a user perspective (e.g., *Storefront UI*).
  - Task / Bug: Technical implementations or defects to resolve.
  - Sub-task: Granular steps within a story or task.
- Product Backlog: A prioritized list of all pending features, enhancements, and bug fixes awaiting assignment.
- Sprint & Scrum Board: A time-boxed cycle (typically 1–2 weeks) where team members pull items from the backlog into active workflow columns (*To Do*, *In Progress*, *In Review*, *Done*).
- Story Points: Relative estimates assessing effort, technical complexity, and risk for each work item.

Jira Setup & Deployment Models

- Jira Cloud (SaaS): Hosted directly by Atlassian. No manual server maintenance or database installations required; teams create an organization site (e.g., company.atlassian.net) and get instant access via a web browser.
- Jira Data Center (Self-Managed): Installed on private infrastructure or dedicated cloud servers (AWS/Azure) for enterprises requiring localized data governance, custom security policies, and high-availability clustering.

DevOps Integration & Automation Lifecycle

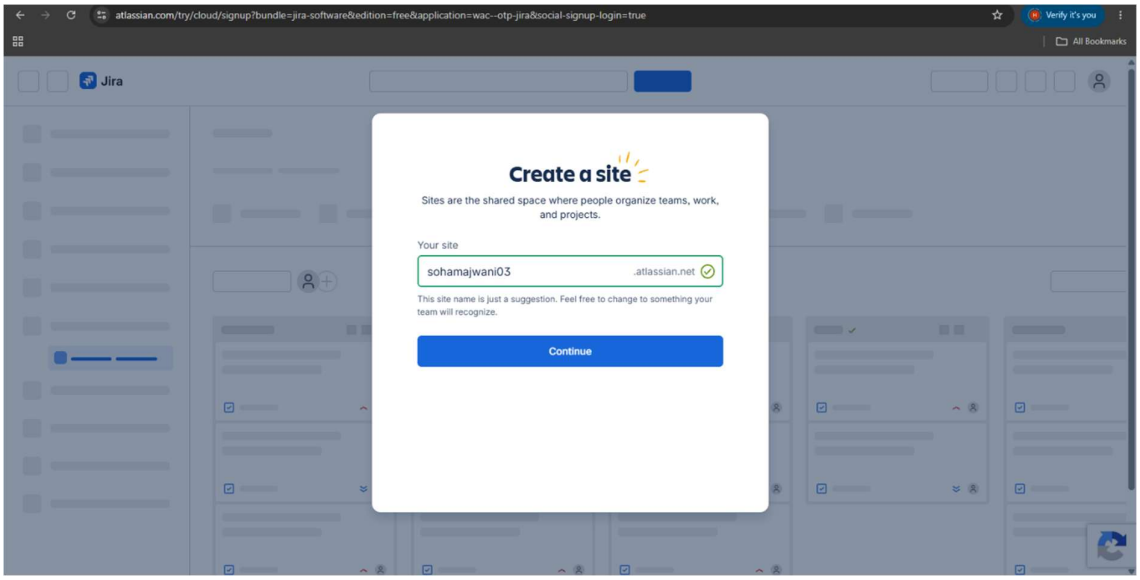
- Version Control Linkage: Integrates with code repositories (such as GitHub, GitLab, or Bitbucket) through official marketplace apps (e.g., *GitHub for Jira*).

- Smart Issue Keys: Mentioning ticket identifiers (e.g., SCRUM-4, SCRUM-6) in branch names, commit messages, or Pull Request titles binds VCS activity directly to the Jira issue panel.
- Jira Automation Rules: Event-driven rules that trigger state transitions across the board (e.g., automatically shifting a ticket to *In Review* when a Pull Request is opened, or transitioning to *Done* upon a merged PR).

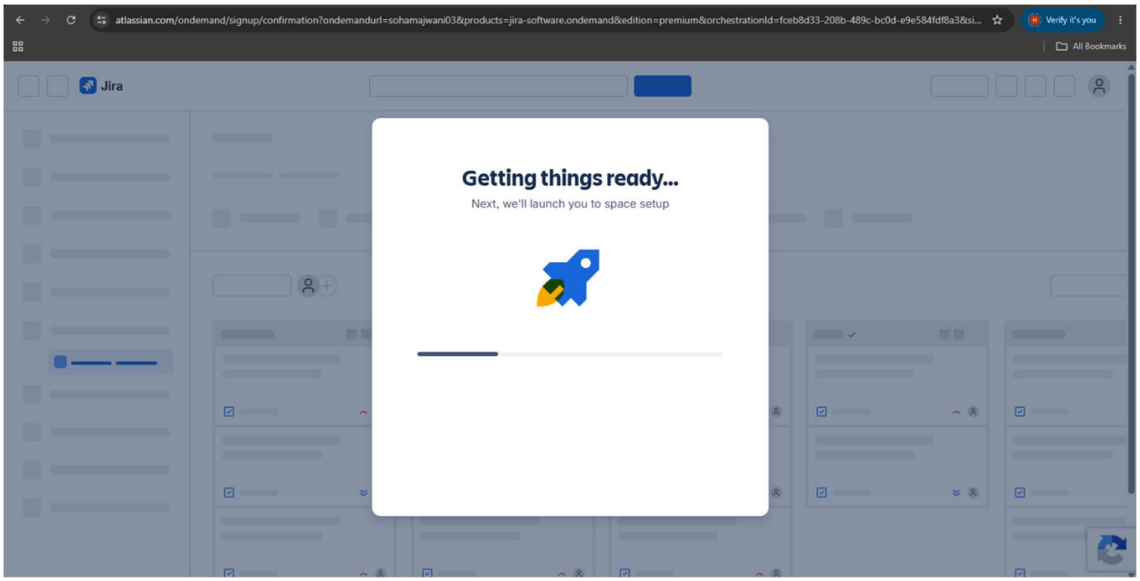
Team Collaboration & Access Management

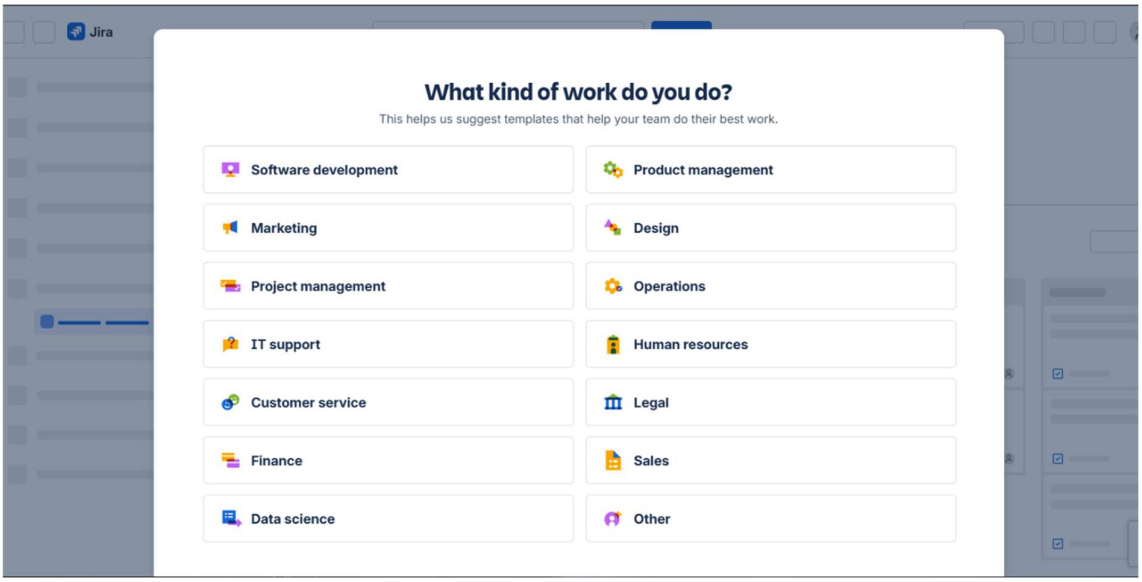
- Role-Based Access Control (RBAC): Access is governed at the organization and project levels via roles (*Administrator*, *Member*, *Viewer*).
- Work Allocation: Tickets are assigned directly to individual developers, allowing visibility into sprint capacity, bottlenecks, and velocity reports.

Output:

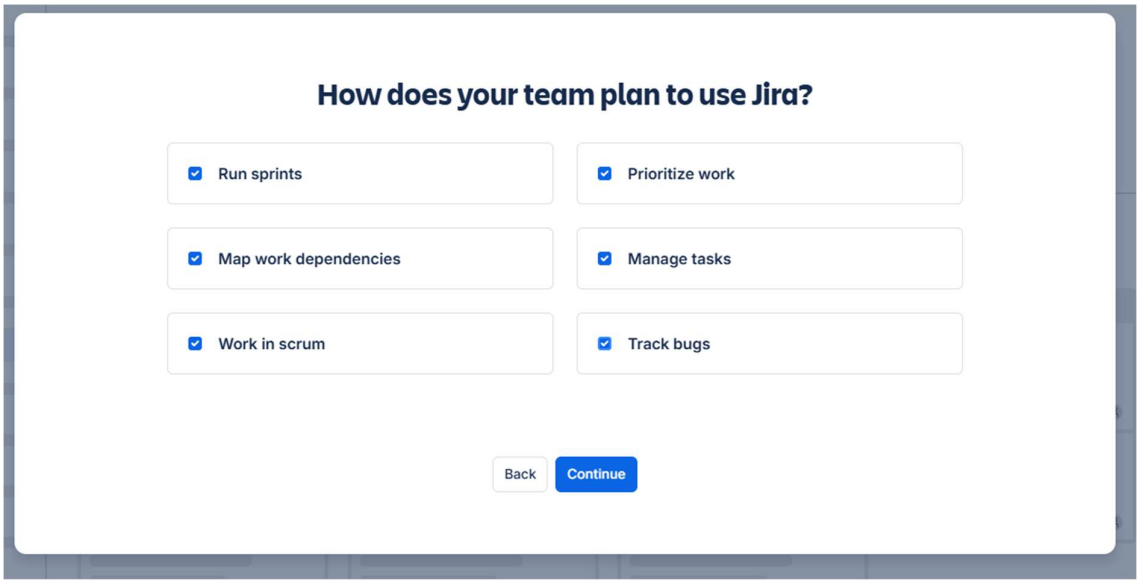


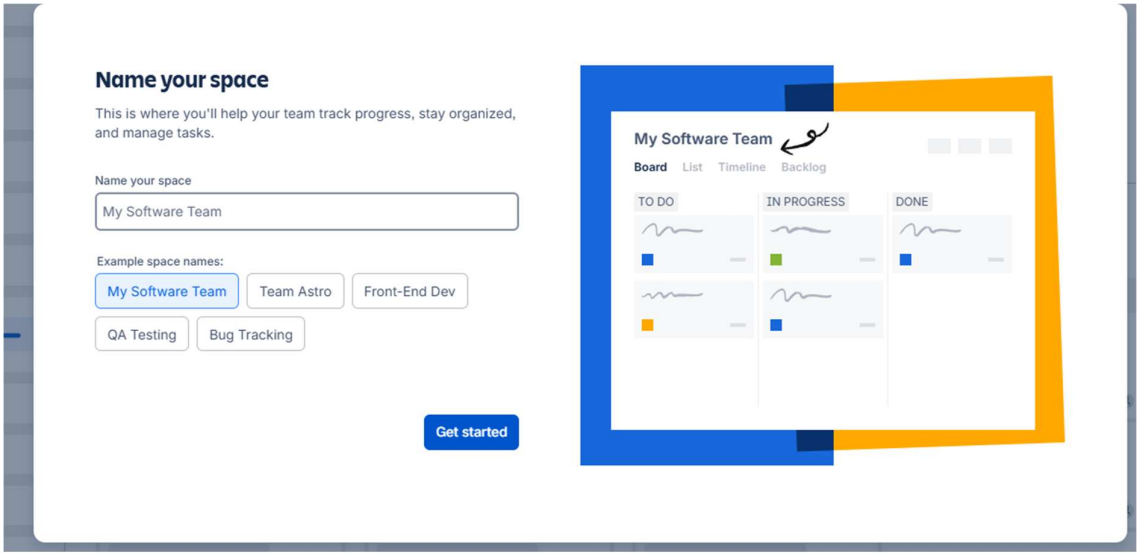
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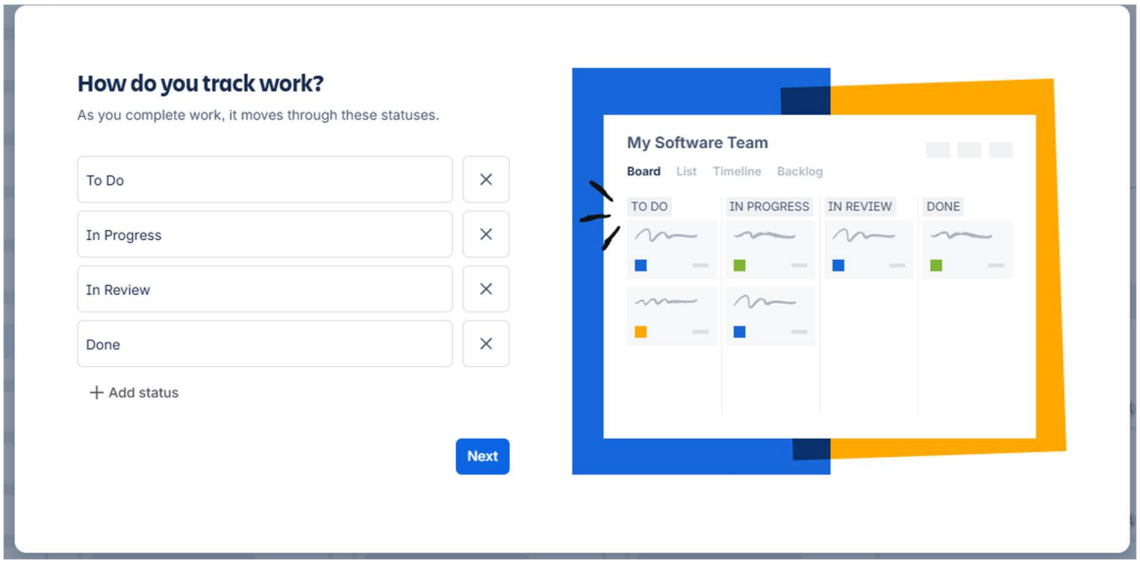


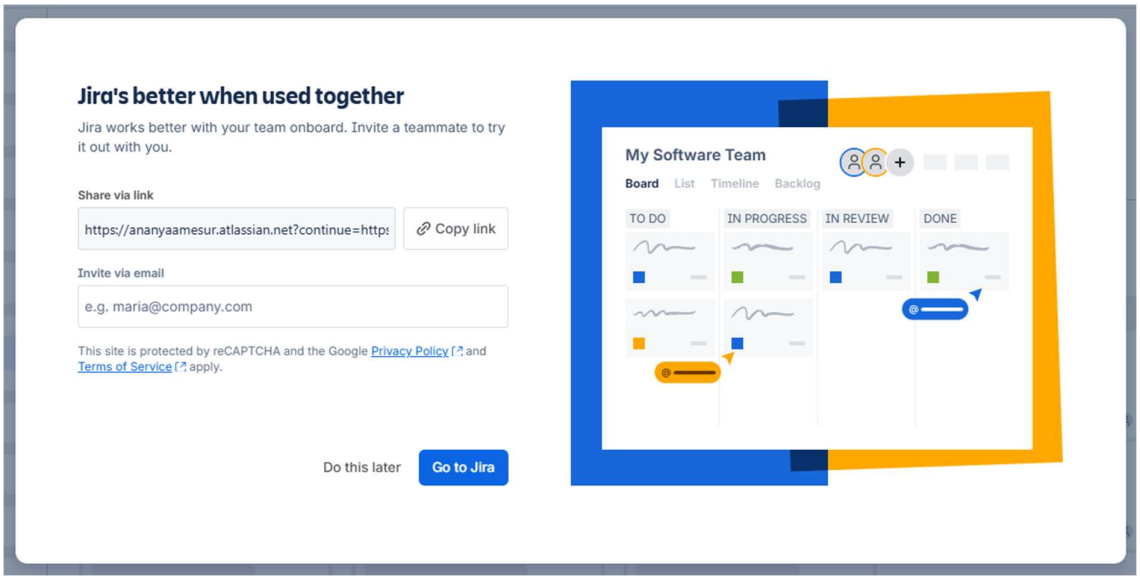
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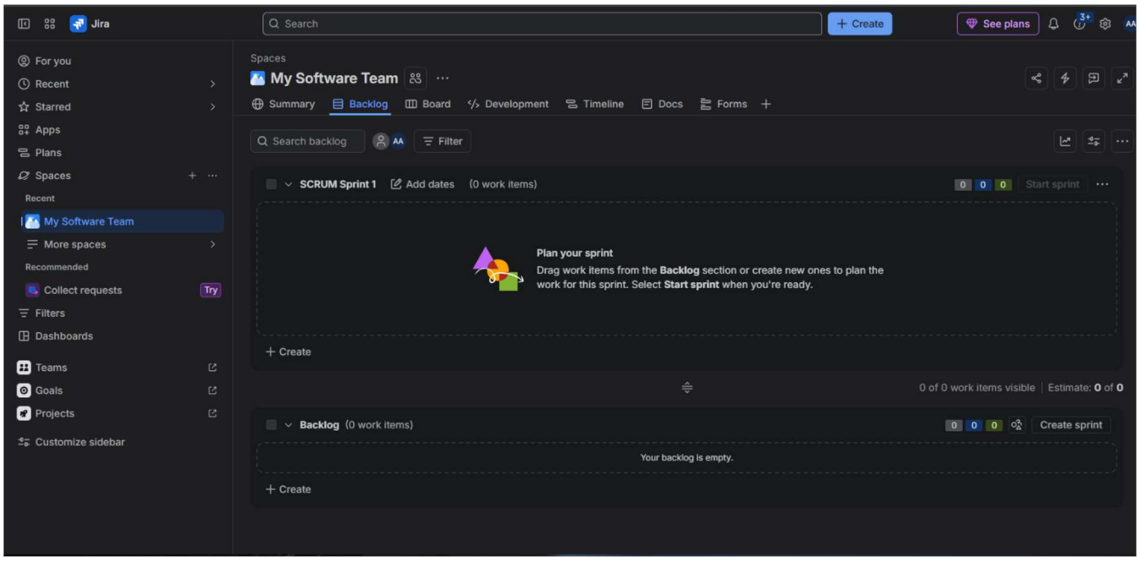


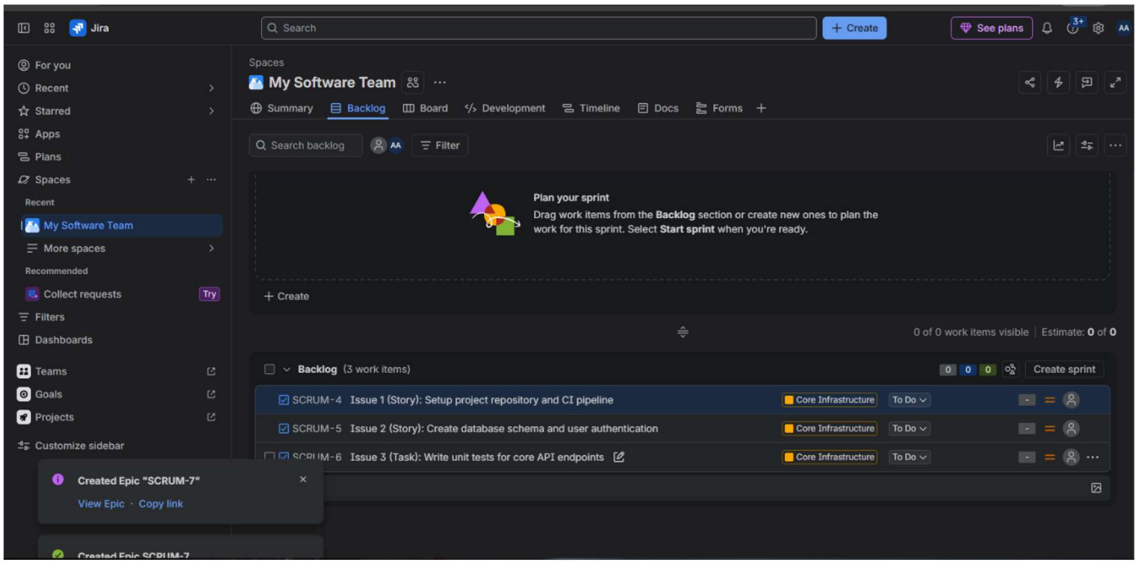
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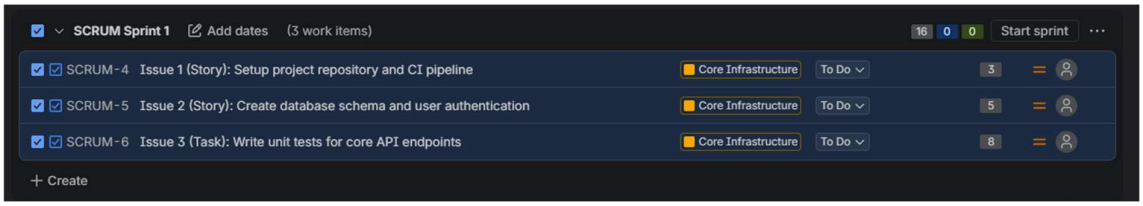




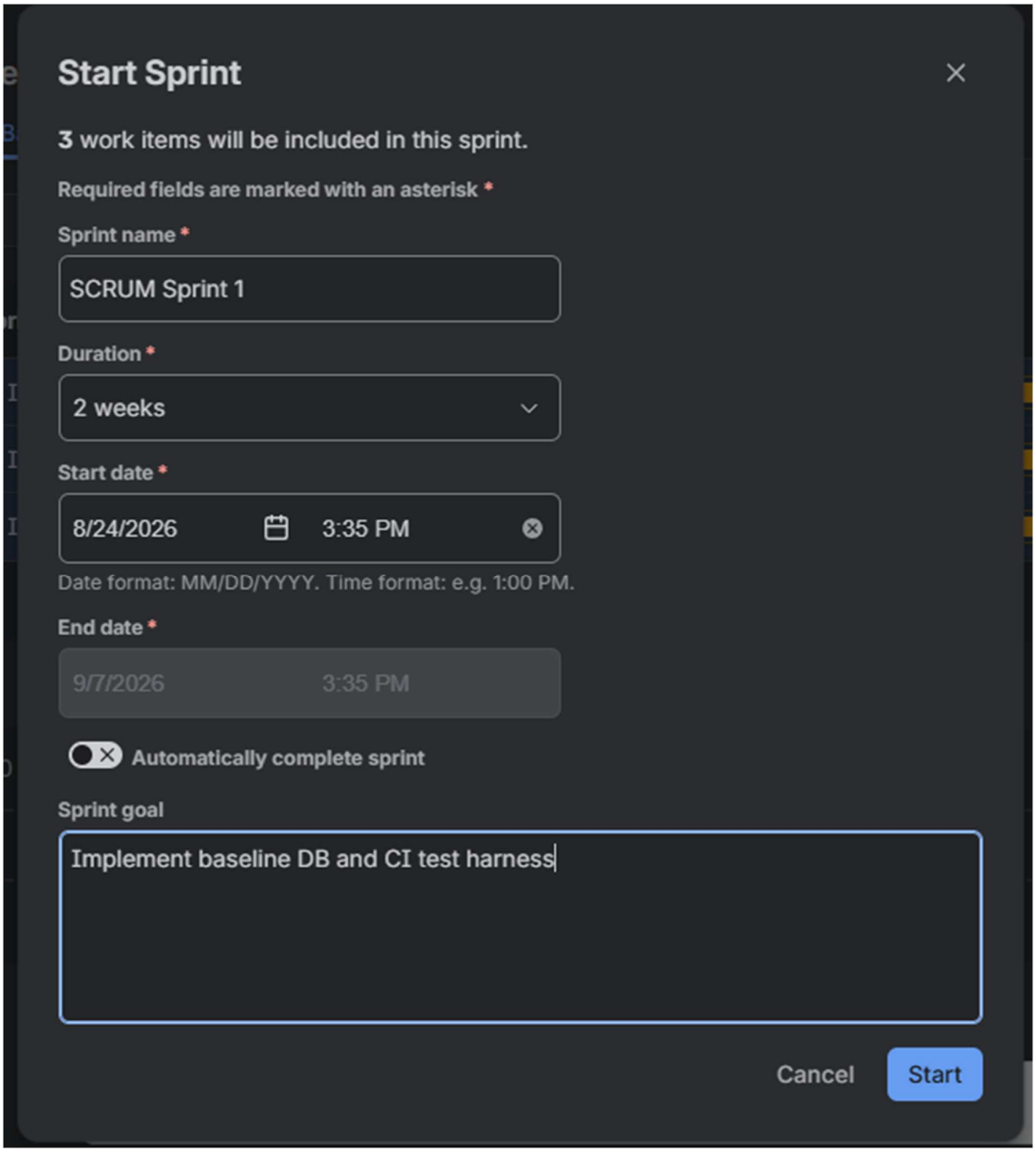
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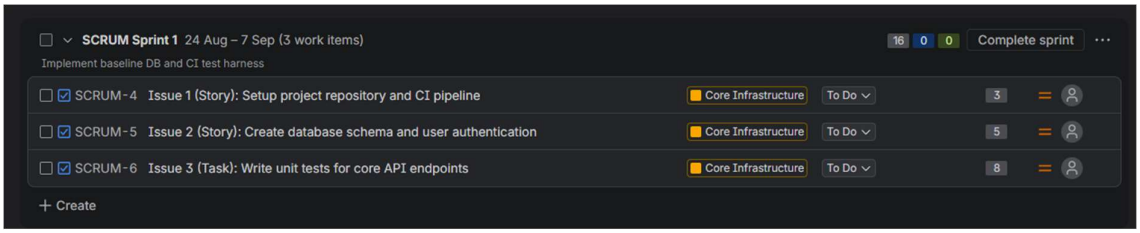




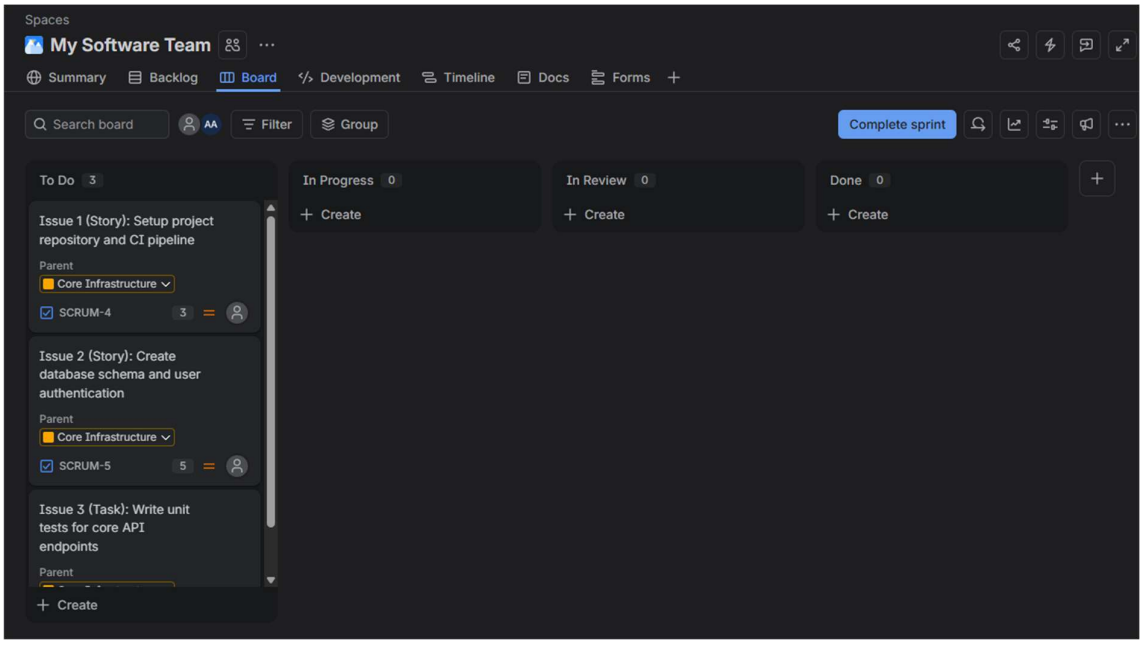


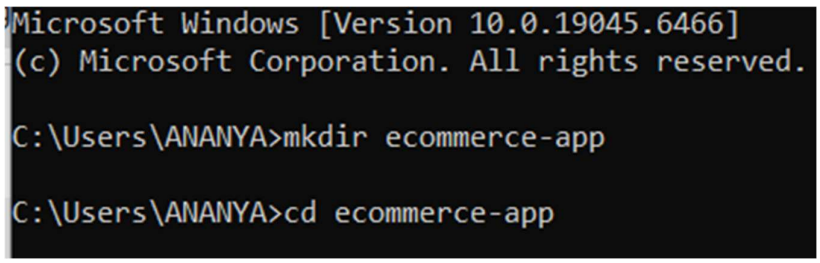
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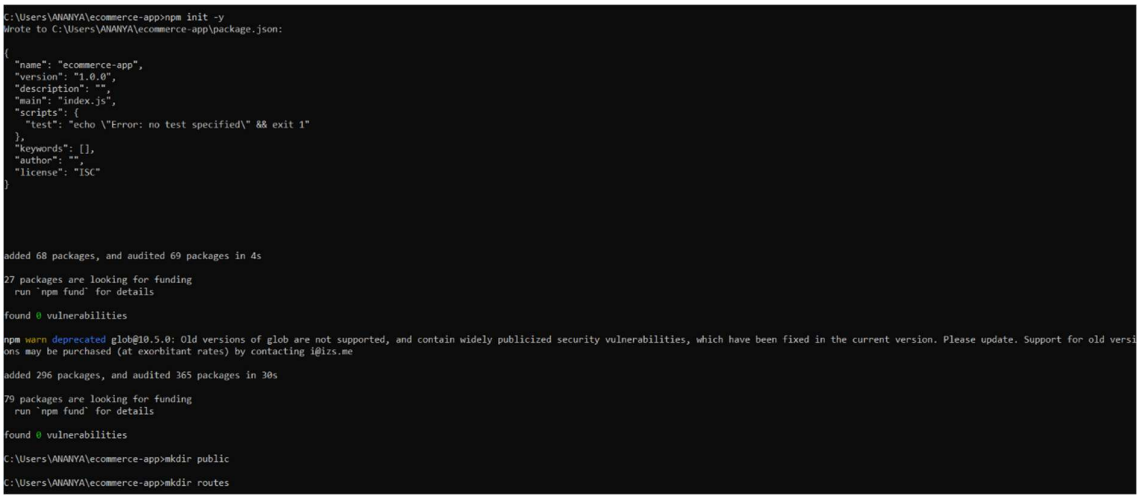




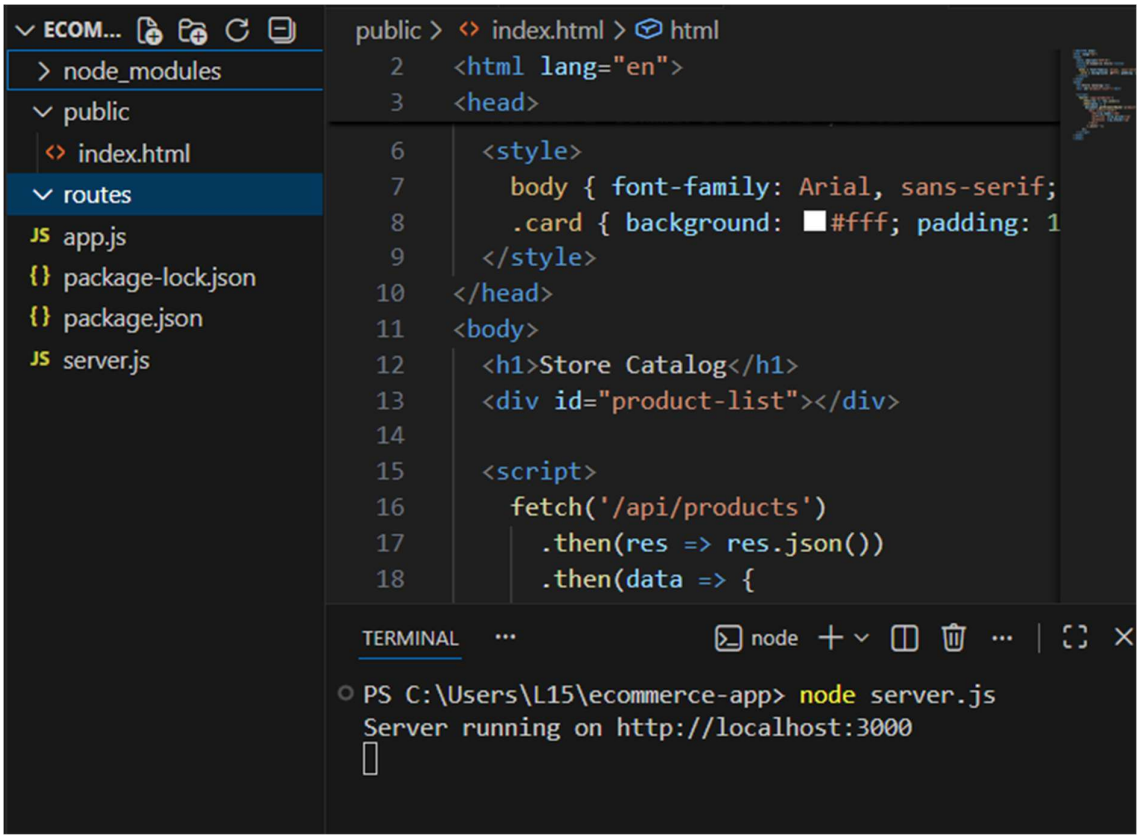
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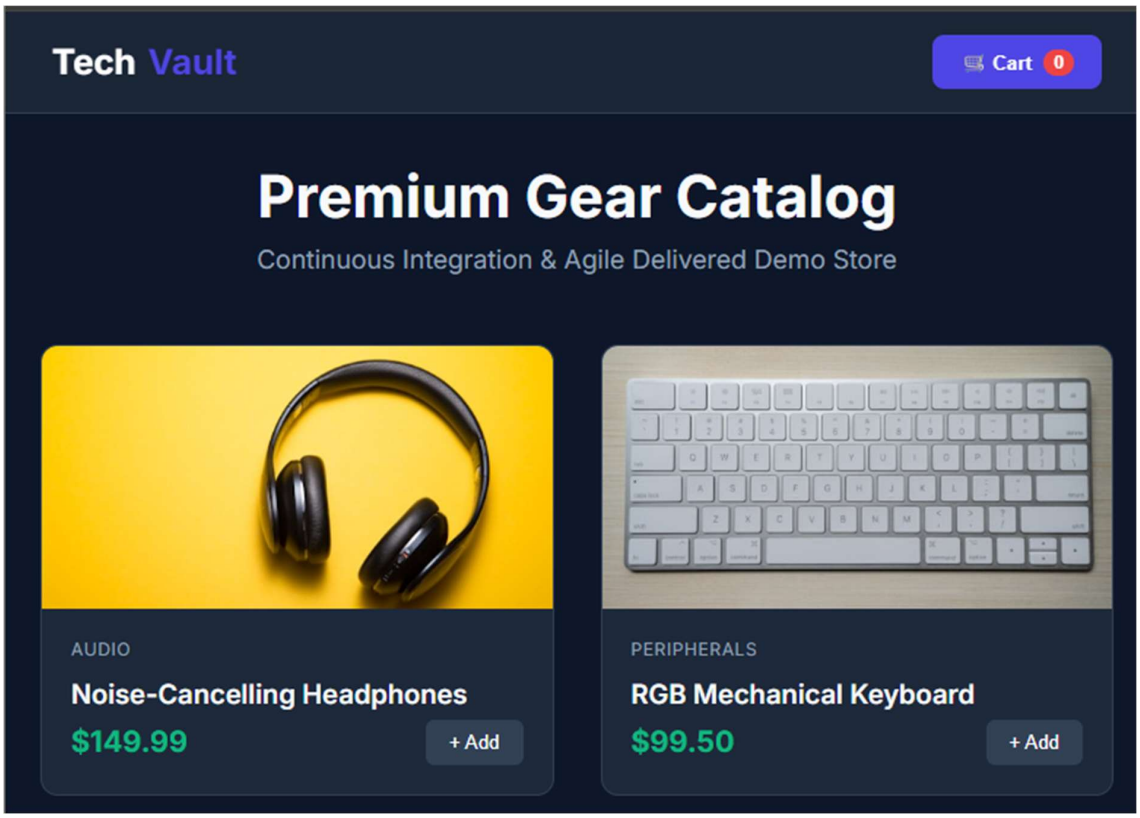




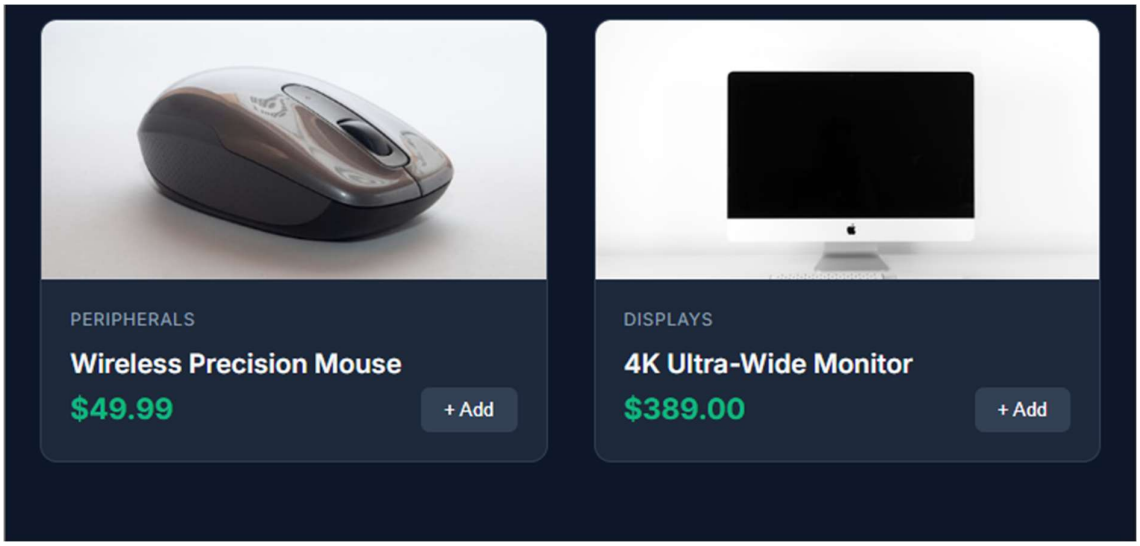


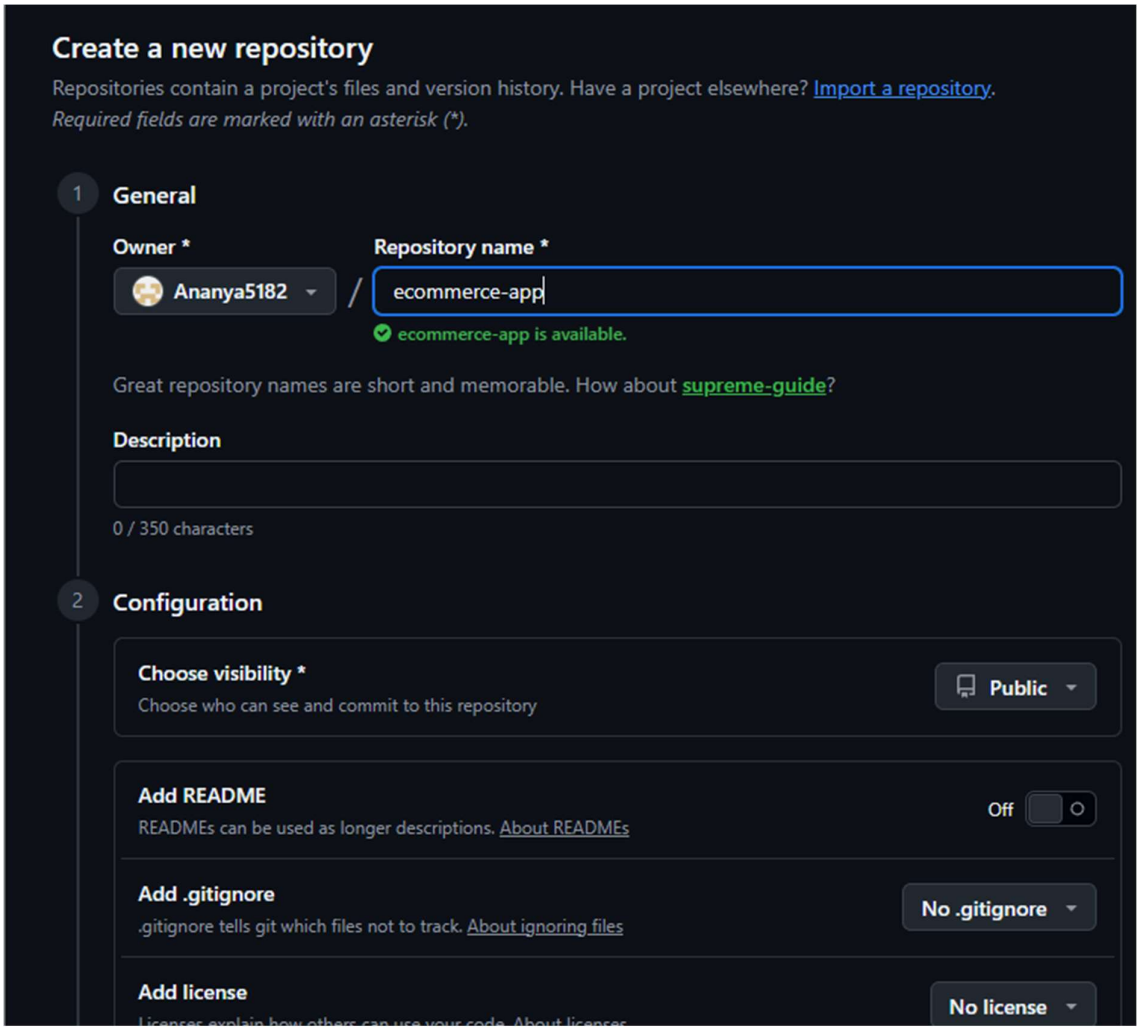
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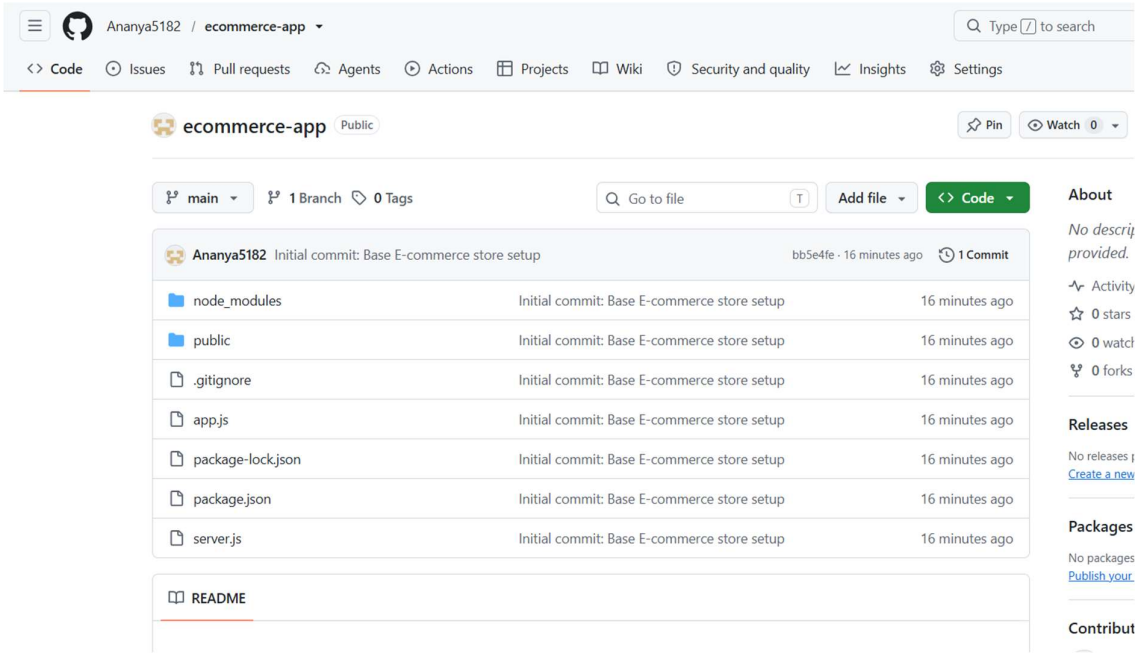


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Collaborators and teams

Public repository
This repository is public and visible to anyone

Manage visibility

Direct access

1 entity has access to this repository. 1 collaborator.

Manage access

Add people

Select all | Type | Find a collaborator...

sohamajwani
Collaborator

Previous Next

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GitHub configuration

Connect a GitHub organization

Connecting GitHub to Atlassian allows you to view GitHub development information on Jira work items, find GitHub work suggestions in the development tab, and more. To see your development data on Jira, your team must include issue keys in branch names, commit messages, and pull request titles.

Even if your organization is still backfilling historical data, you can start using work item keys in your development work immediately.

Connections Settings

GitHub Cloud

Connected repositories

Connected organization	Repository access	Backfill status	Permissions	Settings
Ananya5182	Only select repos 1	FINISHED Backfilled from: 03/03/2026	FULL ACCESS	···

```
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ANANYA\ecommerce-app>git checkout -b SCRUM-5-auth-db-feature
Switched to a new branch 'SCRUM-5-auth-db-feature'

C:\Users\ANANYA\ecommerce-app>git commit --allow-empty -m "SCRUM-5: Connected database schema and user authentication structure"
[SCRUM-5-auth-db-feature 5872924] SCRUM-5: Connected database schema and user authentication structure

C:\Users\ANANYA\ecommerce-app>git push -u origin SCRUM-5-auth-db-feature
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 229 bytes | 229.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'SCRUM-5-auth-db-feature' on GitHub by visiting:
remote:      https://github.com/Ananya5182/ecommerce-app/pull/new/SCRUM-5-auth-db-feature
remote:
To https://github.com/Ananya5182/ecommerce-app.git
 * [new branch]      SCRUM-5-auth-db-feature -> SCRUM-5-auth-db-feature
branch 'SCRUM-5-auth-db-feature' set up to track 'origin/SCRUM-5-auth-db-feature'.

C:\Users\ANANYA\ecommerce-app>
```

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Comparing changes

Choose two branches to see what's changed or to start a new pull request. If you need to, you can also compare across forks or learn more about dif

base: main ← compare: SCRUM-5-auth-db-feature ✓ Able to merge. These branches can be automatically merged.

Add a title *

SCRUM-5: Implement Database Schema and User Authentication.

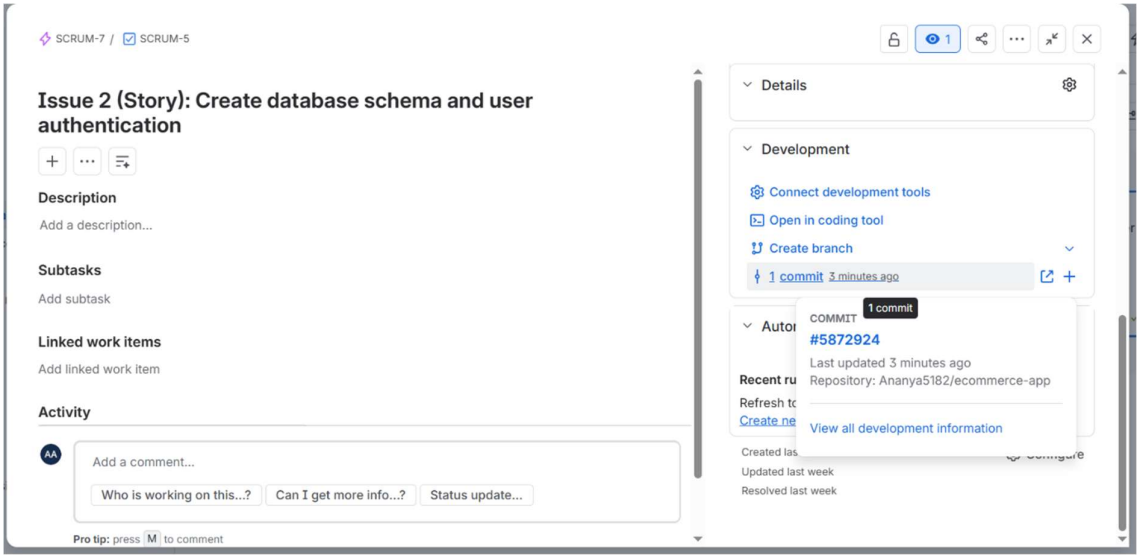
Add a description

Write Preview

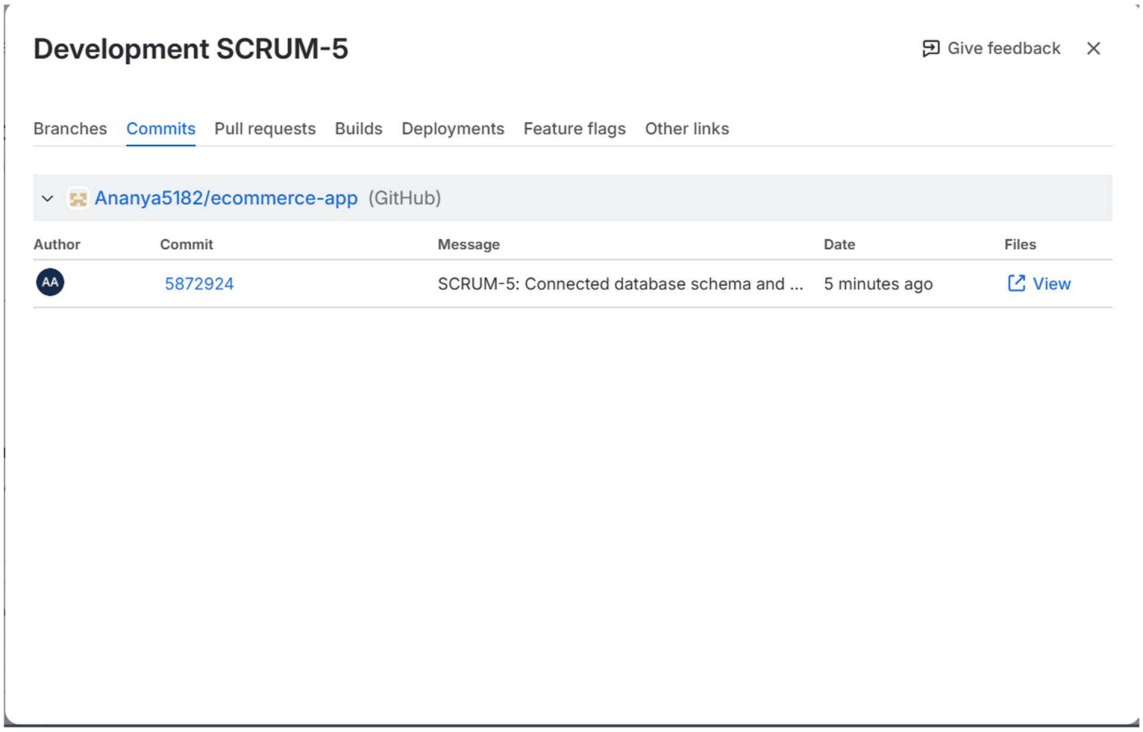
Add your description here...

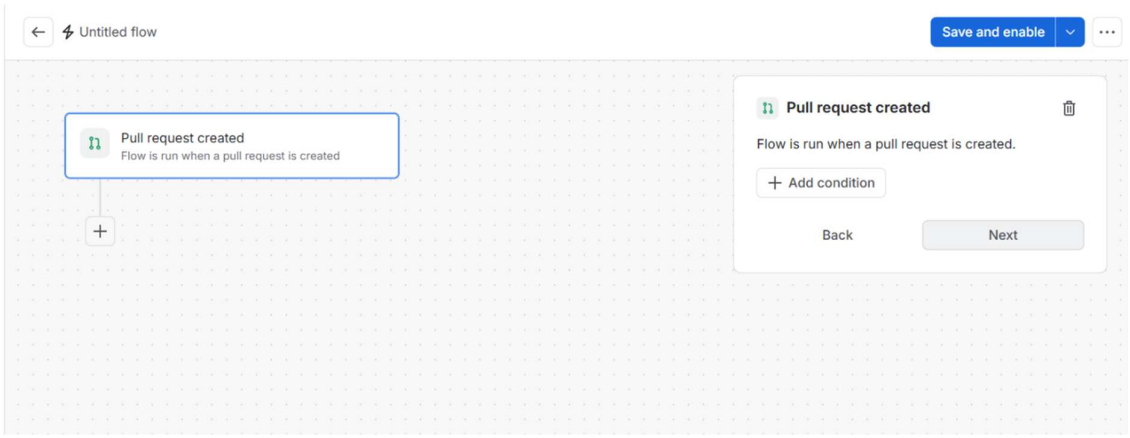
Markdown is supported | Paste, drop, or click to add files

Create pull request

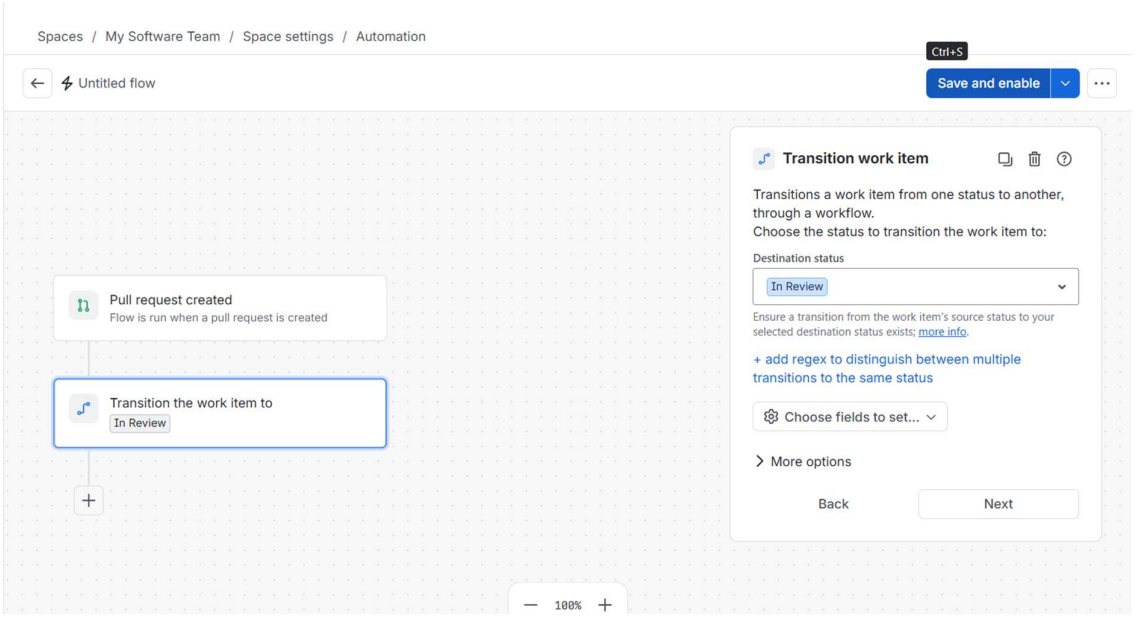


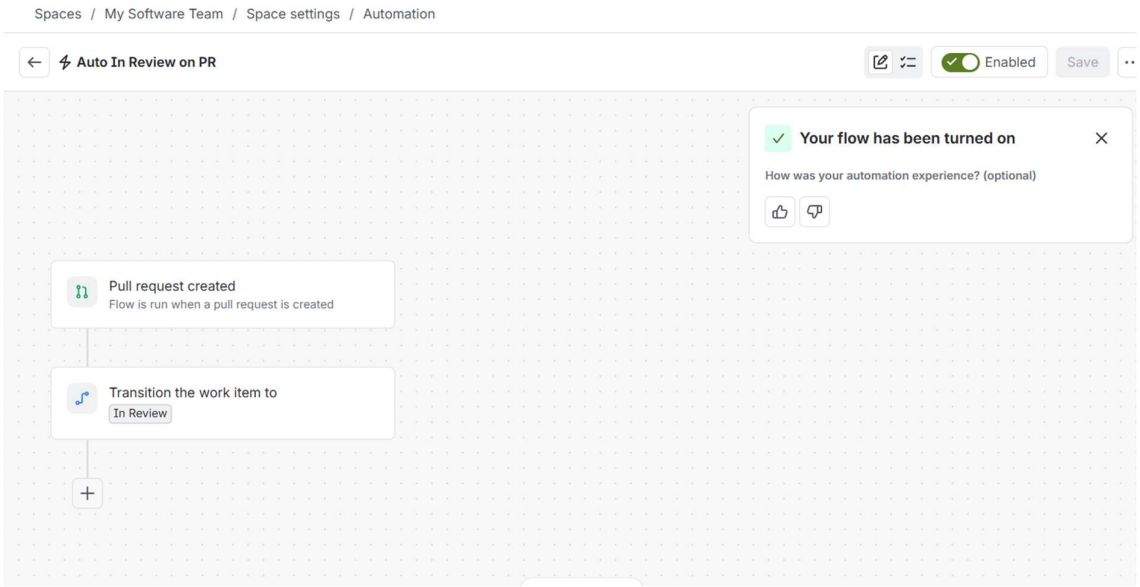
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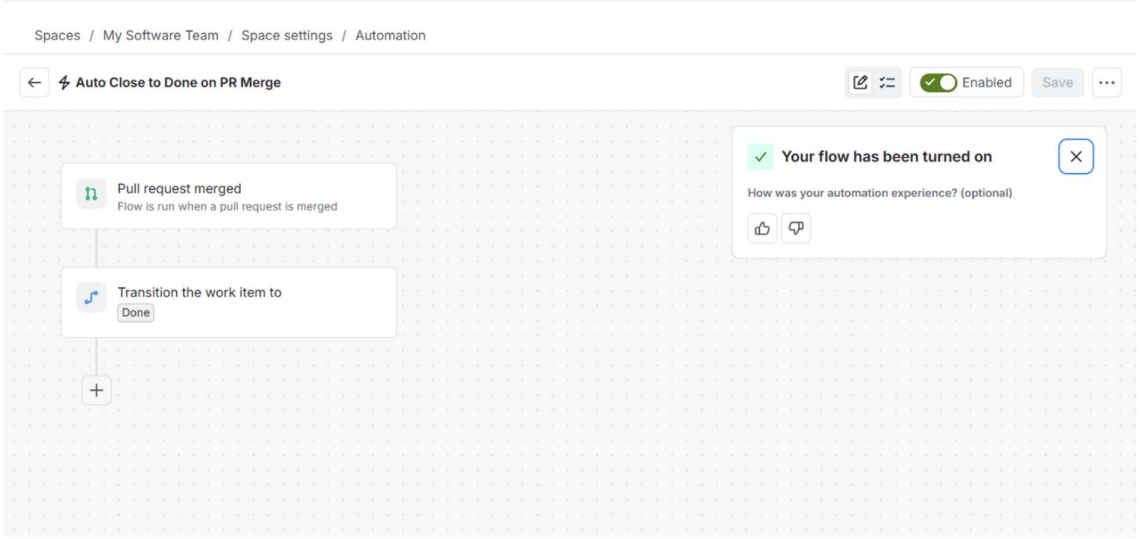


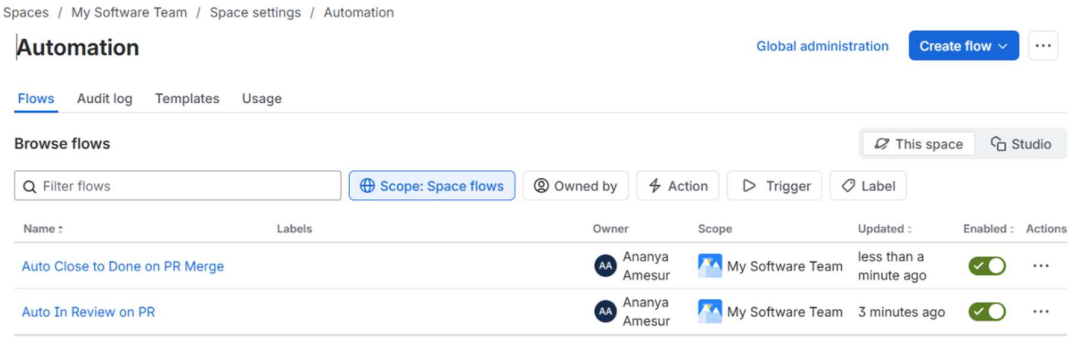
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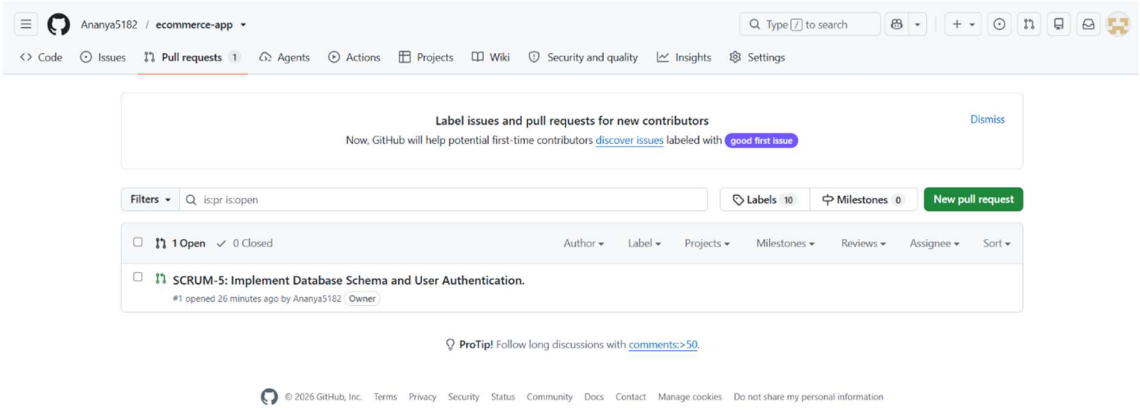




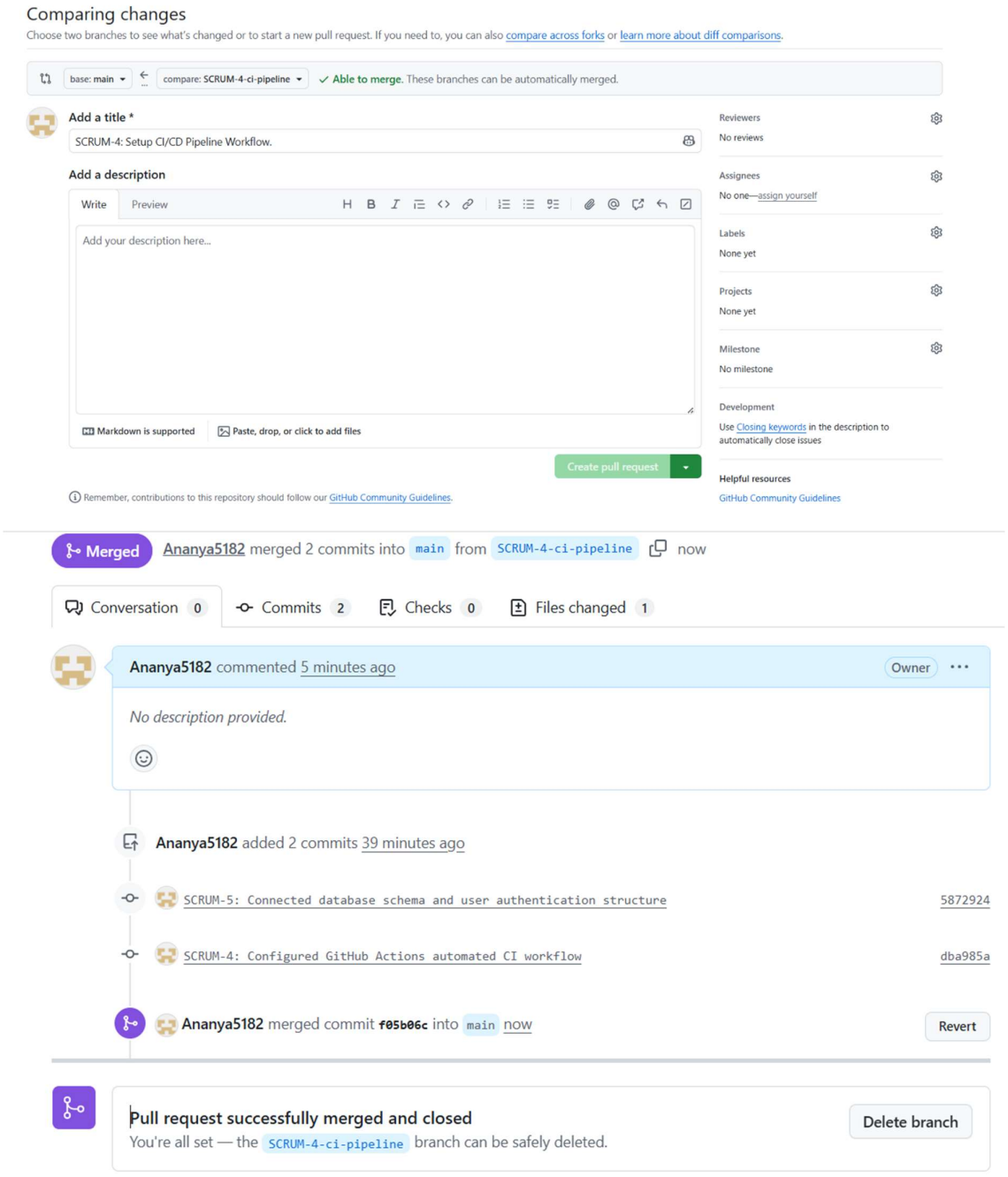
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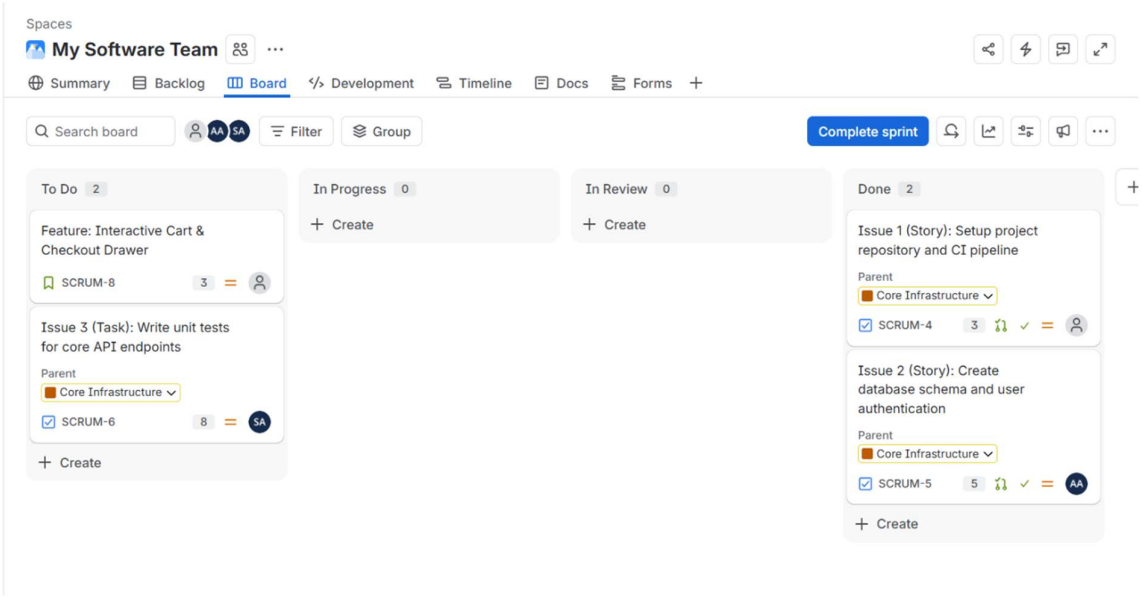


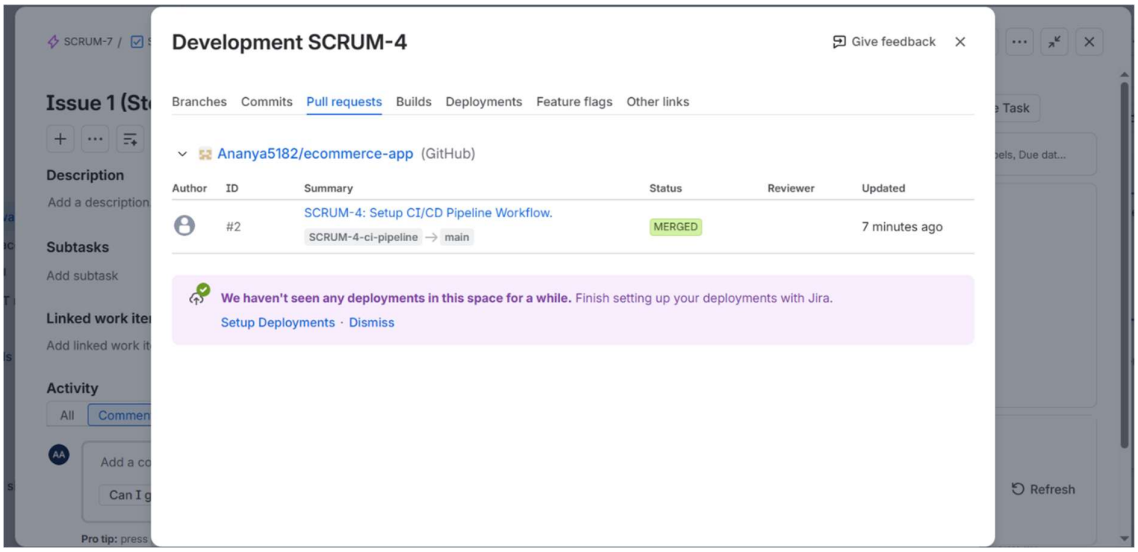


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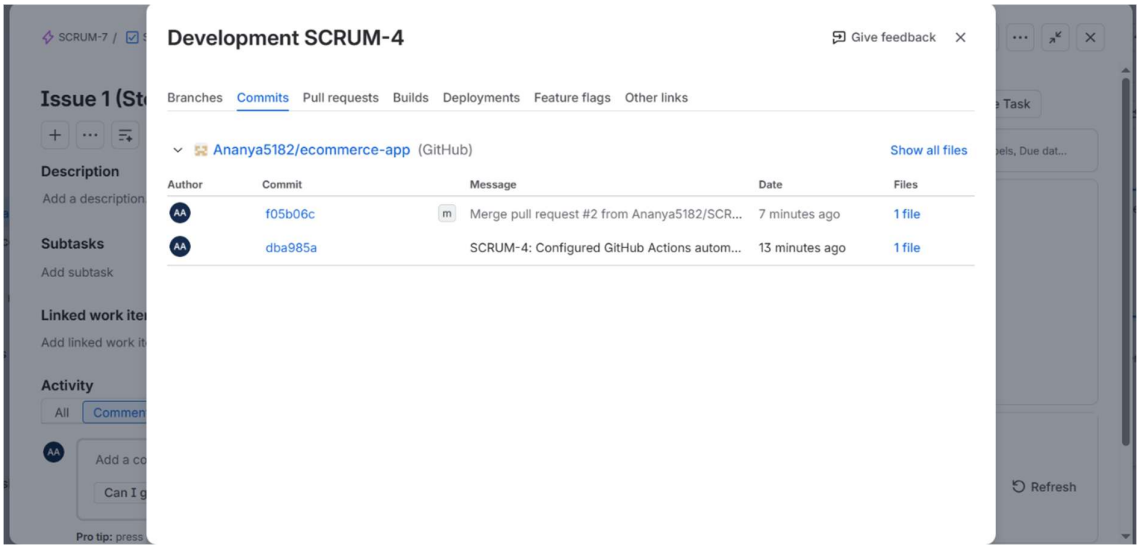


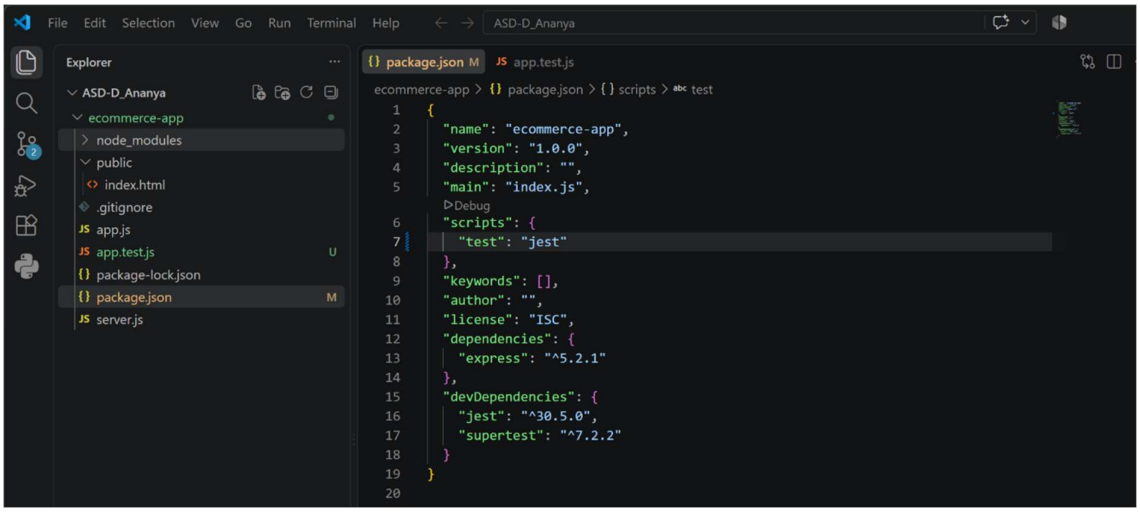
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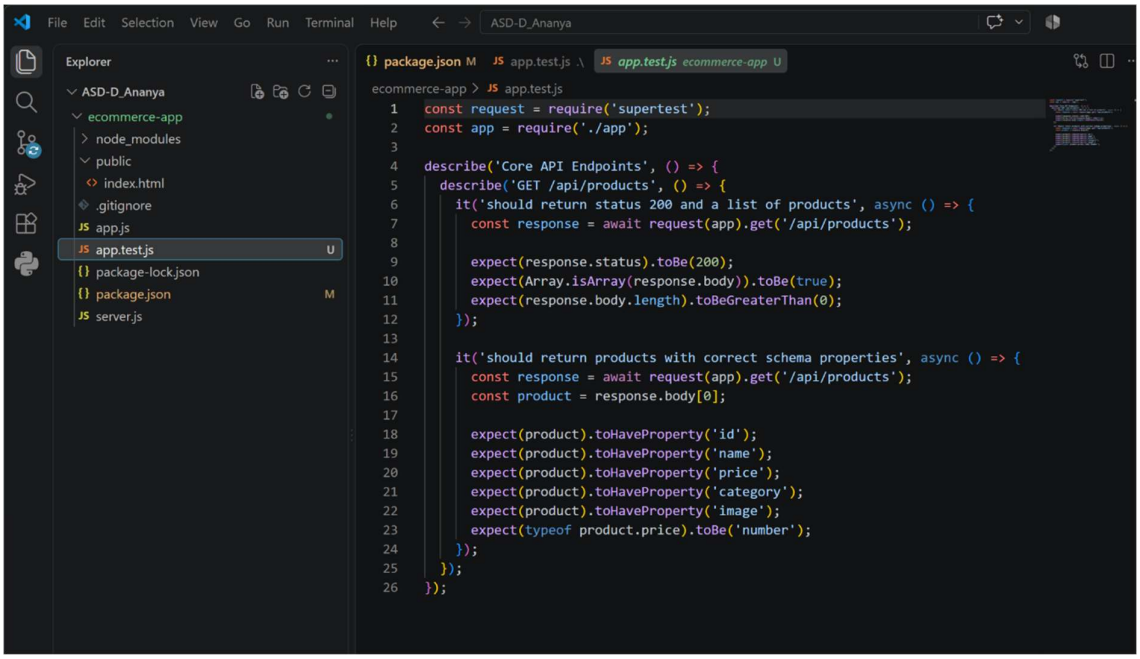


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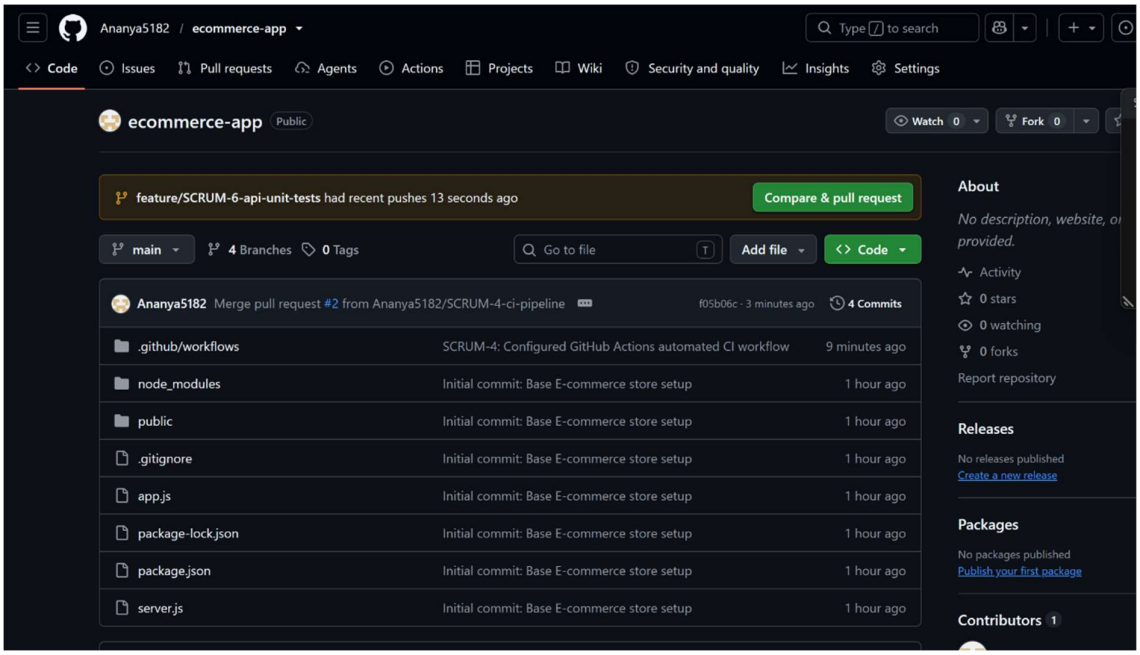

```
PS C:\Users\soham\Documents\ASD-D_Ananya\ecommerce-app> node -v
v24.20.0
PS C:\Users\soham\Documents\ASD-D_Ananya\ecommerce-app> npm -v
11.19.0
PS C:\Users\soham\Documents\ASD-D_Ananya\ecommerce-app> npm test

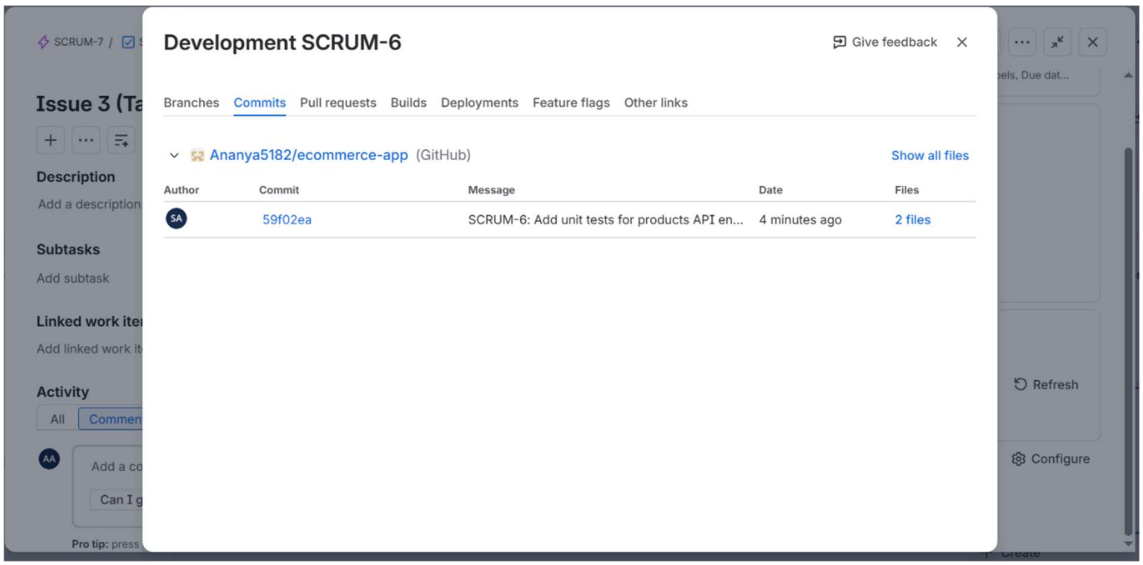
> ecommerce-app@1.0.0 test
> jest

 PASS  ./app.test.js
  Core API Endpoints
    GET /api/products
      √ should return status 200 and a list of products (55 ms)
      √ should return products with correct schema properties (12 ms)

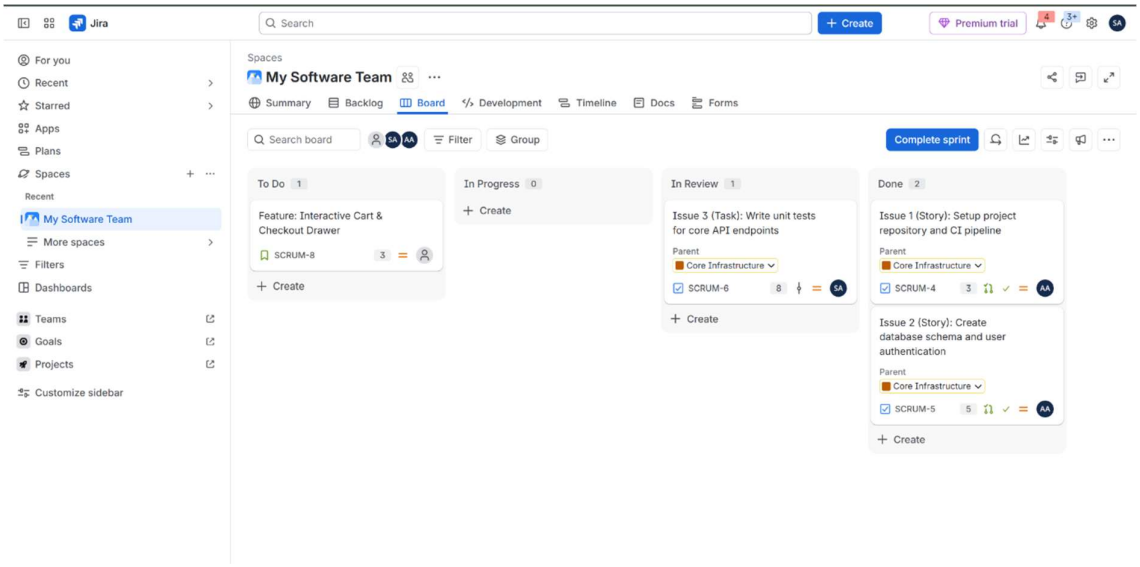
Test Suites: 1 passed, 1 total
Tests:       2 passed, 2 total
Snapshots:   0 total
Time:        1.061 s
Ran all test suites.
npm notice
npm notice New major version of npm available! 11.19.0 -> 12.0.2
npm notice Changelog: https://github.com/npm/cli/releases/tag/v12.0.2
npm notice To update run: npm install -g npm@12.0.2
npm notice
PS C:\Users\soham\Documents\ASD-D_Ananya\ecommerce-app>
```

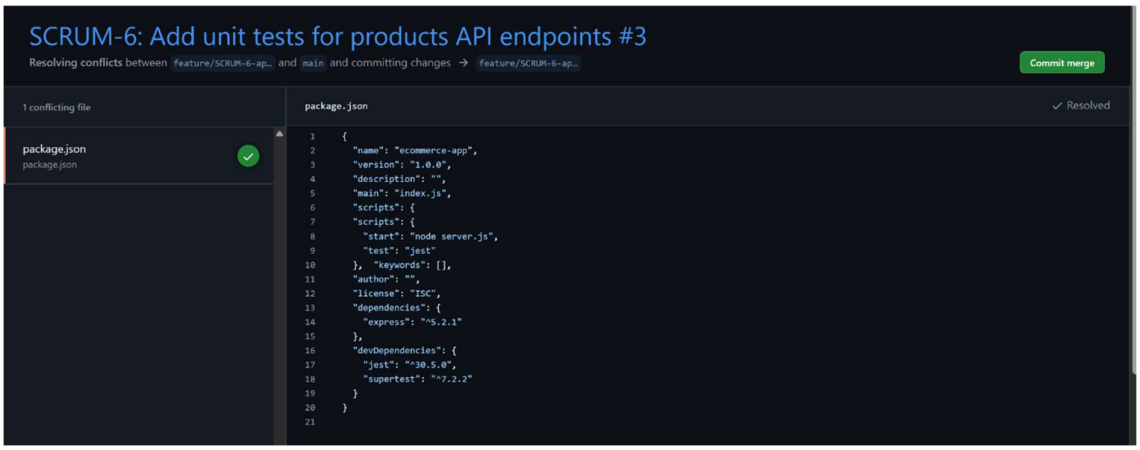
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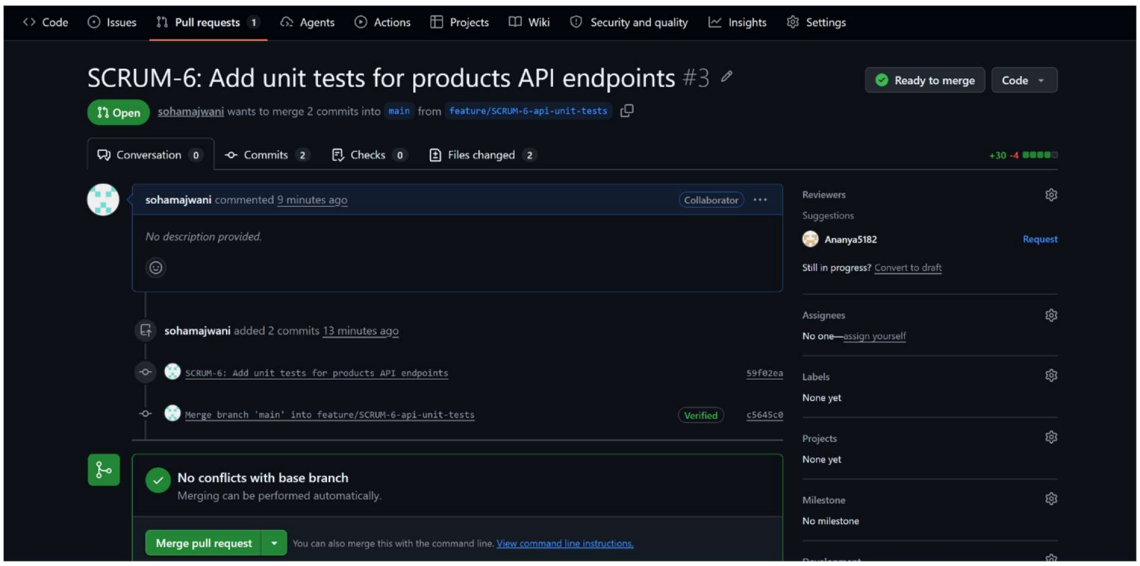




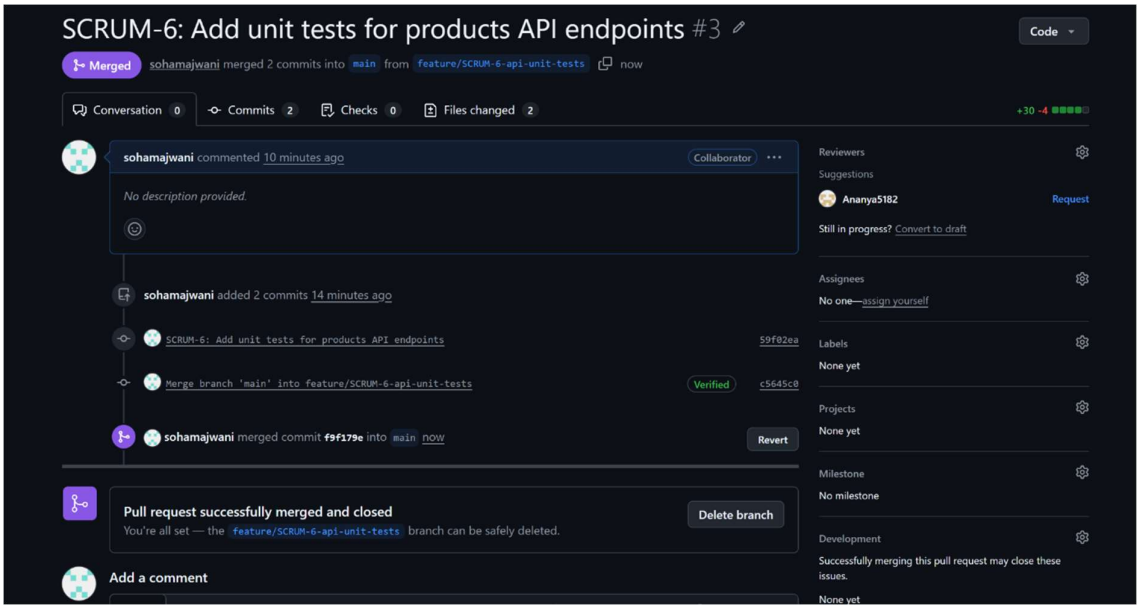
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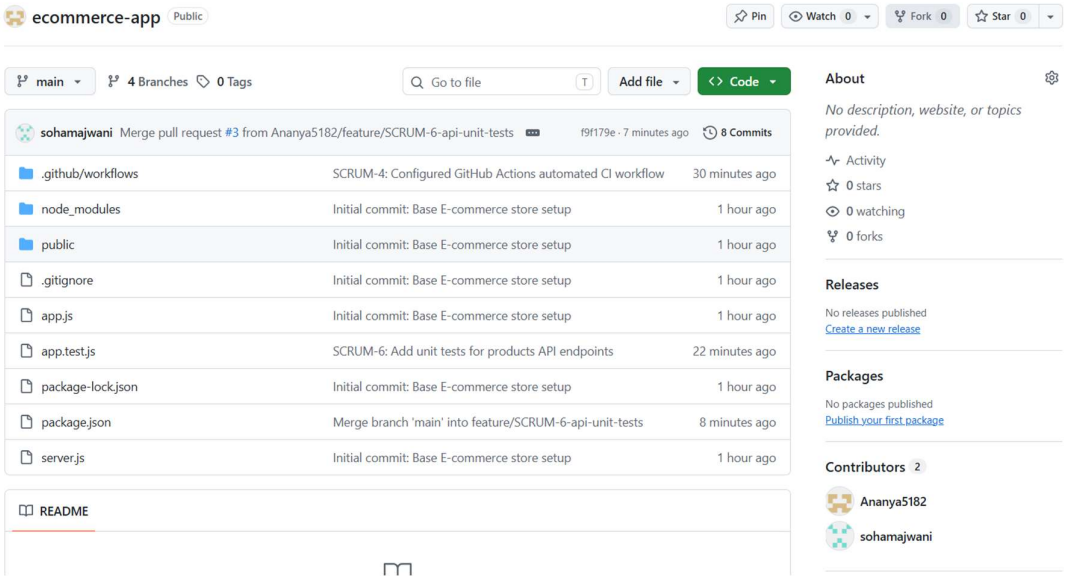




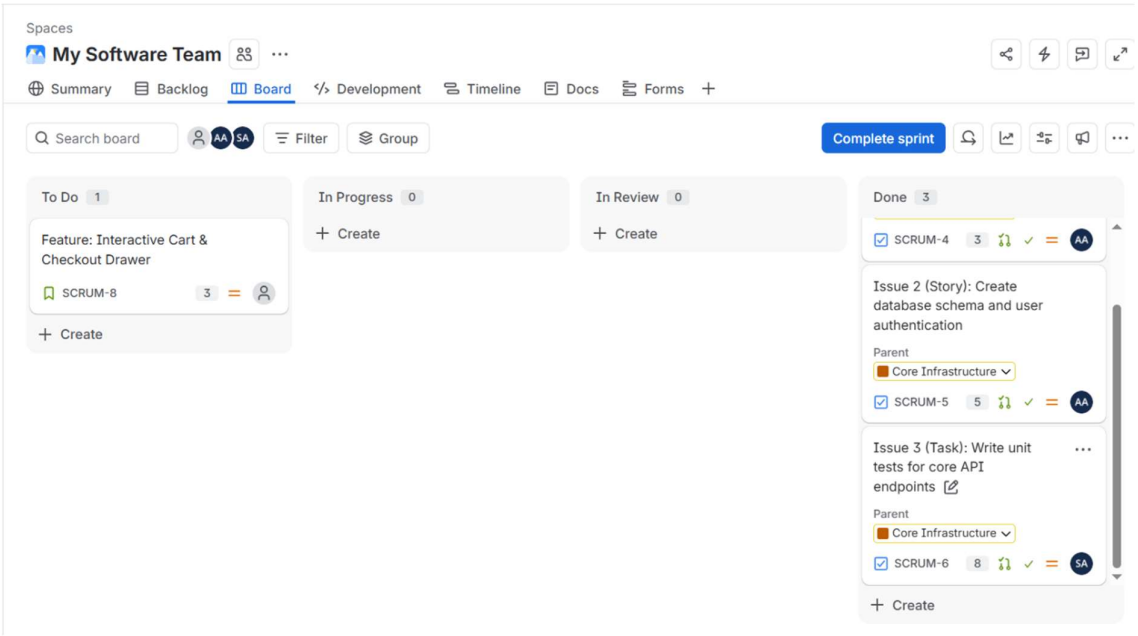


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Conclusion

Integrating Jira with GitHub established an automated, end-to-end Agile development lifecycle for our E-commerce application. We learned how to organize backlogs into active sprints, assign tasks across team members, and track real-time progress using Scrum boards. By referencing issue keys in branch names and Pull Requests, we enabled seamless traceability between our codebase and Jira tickets. Furthermore, setting up GitHub Actions CI pipelines and Jira Automation demonstrated how modern DevOps workflows eliminate manual overhead by automating testing and board transitions upon code merges.